

Microservices with API gateway

Output - Creating Microservices for account and loan

Microservices with API gateway- Creating Microservices for account and loan:

Developed independent Account and Loan microservices using Spring Boot and exposed REST APIs through separate endpoints. Configured different server ports to enable both services to run simultaneously, demonstrating the fundamentals of a microservices architecture.

Step 1: Create Workspace

Step 2: Create Account Microservice

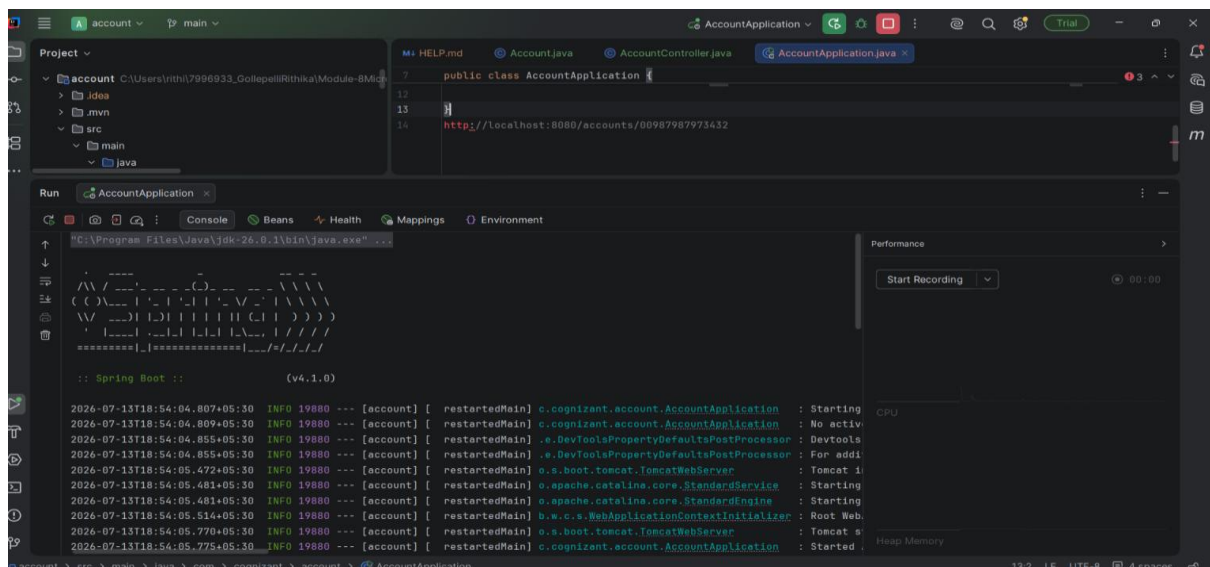
Step 3: Open Project in IntelliJ

Step 4: Build Project

Step 5: Create Model Class

Step 6: Create Controller

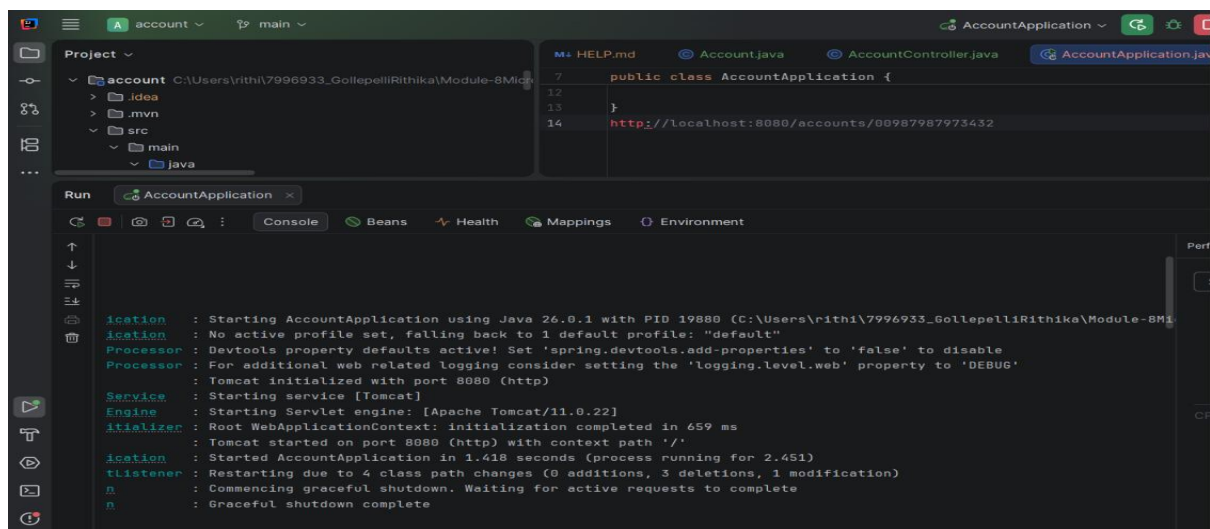
Step 7: Run Application



The screenshot shows the IntelliJ IDEA IDE with the 'AccountApplication.java' file open. The code defines a Spring Boot application with a REST endpoint for accounts. The Run console shows the application starting successfully on port 8080.

```
public class AccountApplication {  
    // ...  
    http://localhost:8080/accounts/00987987973432  
}
```

```
2026-07-13T18:54:04.807+05:30 INFO 19880 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : Starting  
2026-07-13T18:54:04.809+05:30 INFO 19880 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : No activ  
2026-07-13T18:54:04.855+05:30 INFO 19880 --- [account] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools  
2026-07-13T18:54:04.855+05:30 INFO 19880 --- [account] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For addi  
2026-07-13T18:54:05.472+05:30 INFO 19880 --- [account] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat 1  
2026-07-13T18:54:05.481+05:30 INFO 19880 --- [account] [ restartedMain] o.apache.catalina.core.StandardService : Starting  
2026-07-13T18:54:05.481+05:30 INFO 19880 --- [account] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting  
2026-07-13T18:54:05.514+05:30 INFO 19880 --- [account] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root Web  
2026-07-13T18:54:05.770+05:30 INFO 19880 --- [account] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat 1  
2026-07-13T18:54:05.775+05:30 INFO 19880 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : Started
```



The screenshot shows the Run console output for the AccountApplication. It details the startup process, including the initialization of the Tomcat web server and the application's readiness to serve requests on port 8080.

```
ication : Starting AccountApplication using Java 26.0.1 with PID 19880 (C:\Users\rithi\7996933_GollepelliRithika\Module-8Mi  
ication : No active profile set, falling back to 1 default profile: "default"  
Processor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false' to disable  
Processor : For additional web related logging consider setting the 'logging.level.web' property to 'DEBUG'  
Service : Tomcat initialized with port 8080 (http)  
Service : Starting service [Tomcat]  
Engine : Starting Servlet engine: [Apache Tomcat/11.0.22]  
itializer : Root WebApplicationContext: initialization completed in 659 ms  
ication : Tomcat started on port 8080 (http) with context path '/'  
ication : Started AccountApplication in 1.418 seconds (process running for 2.451)  
tlistener : Restarting due to 4 class path changes (0 additions, 3 deletions, 1 modification)  
n : Commencing graceful shutdown. Waiting for active requests to complete  
n : Graceful shutdown complete
```

Step 8: Test in Browser

Account Microservice is working perfectly.

<http://localhost:8080/accounts/00987987973432>

The response in my browser matches the expected output is :



Step 9: Create Loan Microservice

Step 10: Open Loan Project

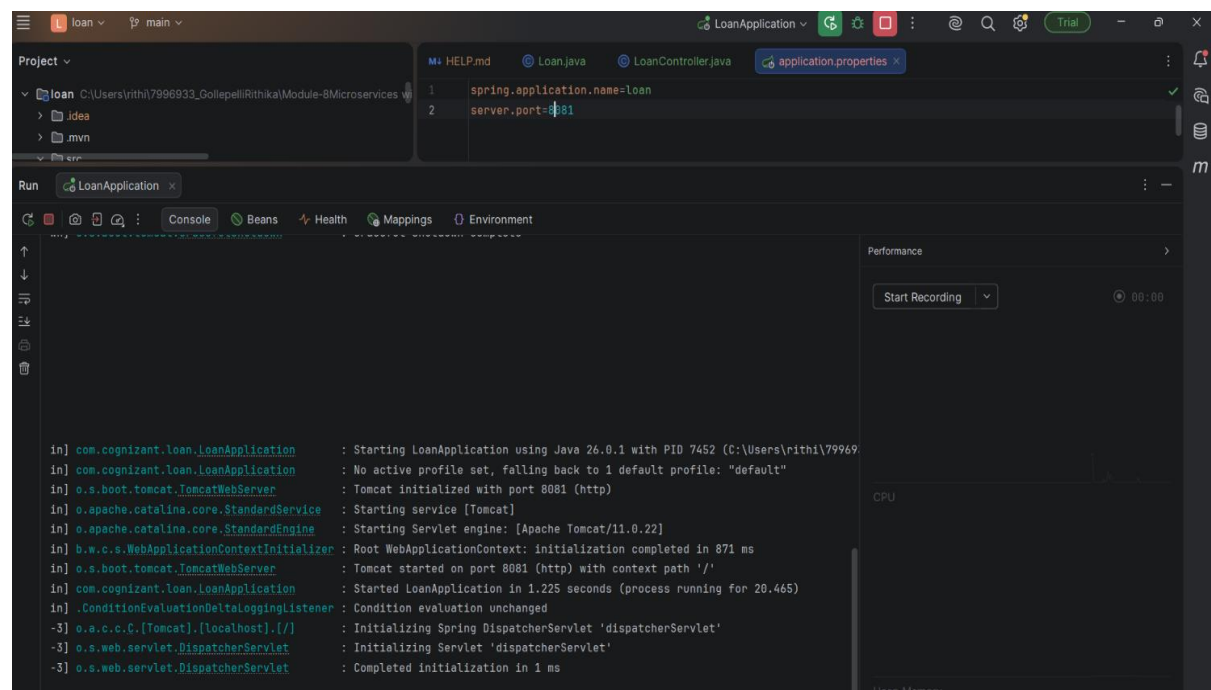
Step 11: Build- Terminal-mvn clean package

Step 12: Create Loan Model

Step 13: Create Loan Controller

Step 14: Change Port

Step 15: Run Loan Application



Step 16: Test

Open browser

<http://localhost:8081/loans/H00987987972342>



- ☐ **Account** service running in IntelliJ (console showing port **8080**).
- ☐ **Loan** service running in IntelliJ (console showing port **8081**).
- ☐ Browser output for:
 - <http://localhost:8080/accounts/00987987973432>
 - <http://localhost:8081/loans/H00987987972342>